

Effects of using the Informed Health Choices digital secondary school resources on the ability of Rwandan students to think critically about health: protocol for a cluster-randomised trial

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Colophon

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Abstract

Background: Young people are exposed to many claims about the effects of things they can do to protect or improve health. To make good health choices, they need the ability to assess the reliability of those claims critically. Critical thinking is a core competence in the Rwandan secondary school curriculum and in many countries. However, critical thinking about health is rarely taught in Rwanda. Digital secondary school resources for critical thinking about health can potentially address this gap and if published be widely disseminated at low cost. The objective of the planned study is to evaluate the effects of using digital secondary school resources to help students learn to think critically about health.

Methods: We will conduct a two-arm cluster-randomized trial. We will randomly select 84 lower secondary schools from 10 districts representing all five provinces of Rwanda. Using stratified random allocation, we will assign 84 schools to the intervention or control arm. Schools in the intervention arm will teach 10-lessons after teacher training on the content of the secondary school resources. Schools in the control arm will not receive any training and resources. They will carry on with teaching the national curriculum. The primary outcome will be the proportion of students with a predetermined passing score on a test with multiple-choice questions from the Claim Evaluation Tools item bank. The questions measure the ability to apply key concepts related to assessing health claims and making informed health choices. The test will include two questions addressing each of the nine concepts covered by the secondary school resources.

Discussion: This is one of three randomized trials to assess the effects of the Informed Health Choices secondary school resources in Rwanda, Kenya, and Uganda. The findings will inform decisions about how to promote critical thinking about health. **Trial registration:** Pan African Clinical Trial Registry, trial identifier: PACTR202203880375077, Registered on 15th February 2022.

Background

Adolescence is a critical stage in life where young people start to make choices on their own, including health choices. Young people and adults are confronted with health information and claims about what they should and should not do to improve their health. The claims come from elders, peers, others in the community, social media, news, other sources, and many of the claims are unreliable [1–3]. Young people and many adults are unable to assess the trustworthiness of health claims [4]. They sometimes believe and act on unreliable claims, or fail to believe and act on reliable claims, which can result in waste and unnecessary suffering. For example, in some settings, some young people believe that they can use toothpaste to treat acne or use herbal or home-made treatments for conditions such as colds, cough, stomach ache, and injuries [5].

Most health education in Rwandan secondary schools focuses on teaching students to make specific choices, for example, about exercise, what they eat, benefits and harms of alcohol and drugs, or sexual behaviours. Secondary school education rarely focuses on teaching students to think critically about that information or other health choices.

A systematic review of the effectiveness of interventions to teach people key concepts required to assess claims about the effects of health interventions found that well designed educational interventions can improve people's ability to apply such concepts [6]. Another systematic review of the effects of school-based educational interventions to teach adolescents to appraise health claims critically found that school-based interventions may have an effect on knowledge and skills required for critical appraisal of health claims [7]. Neither review found studies that measured long-term effects and most studies in the two reviews were conducted in high-income settings.

The Informed Health Choices (IHC) Network is group of researchers that help the public to learn critical thinking about health, with a focus on low-income countries. We have developed and tested resources for primary school children and a podcast for their parents. We conducted a cluster randomised trial of the primary school intervention in Uganda and found that the resources helped children to assess the reliability of claims about treatment effects [8]. Children who participated in the Ugandan trial retained what they learned after one year [9].

The podcast for the parents of primary school children was also evaluated in Uganda [10]. Although the primary school resources were highly effective, it was hard to scale up use of the resources due to printing costs [11]. Moreover, the content of the resources was not integrated into the curriculum or assessed in official exams.

Prior to this planned trial, we conducted a context analysis in Rwanda to explore how we can overcome barriers to wide use of the secondary school resources that we will test in this trial [5]. Most secondary schools in Rwanda now have “smart classrooms” with computers for students and an Internet connection. Making the resources digital rather than relying on printing will help ensure that they can be used widely at a low cost if they are shown to be effective.

We have developed the resources in close collaboration with teachers, students, and other key stakeholders, including curriculum developers [12]. This will help to ensure that the resources are well suited to the context in which they will be used and integrated in the lessons taught in the Rwandan curriculum. The resources will focus on nine key concepts (Box 1) that were prioritised with curriculum developers and teachers in Rwanda, Kenya, and Uganda [13, 14].

Box 1: Prioritised IHC Key Concepts for the IHC secondary school intervention

Prioritised concepts	
1	Treatments can cause harms as well as benefits.
2	Large, dramatic effects are rare.
3	Personal experiences or anecdotes alone are an unreliable basis for most claims.
4	Treatments that are new or technologically impressive may not be better than available alternatives.
5	Widely used treatments or those that have been used for decades are not necessarily beneficial or safe.
6	Identifying effects of treatments depends on making comparisons.
7	Small studies may be misleading.
8	Comparison groups should be as similar as possible.
9	Weigh the benefits and savings against the harms and costs of acting or not.

There are several reasons for teaching critical thinking about health in secondary schools. Young people are eager to learn, and adapt easily [15]. Critical thinking is among the key competences that many countries, including Rwanda, have included in their primary and secondary school curricula [5, 16]. Young people are exposed to health information but lack necessary skills to think critically about that information and make well-informed choices. Health is important to everyone, and it is necessary to understand and apply key concepts to assess the reliability of claims about health. Key concepts that are necessary for assessing the reliability of claims and making decisions about other types of

interventions are largely the same [17]. Lastly, young people make up 16% of the world population, and 50% of Rwanda's population is under 20 years-old [18]. Investing in their health education, specifically improving their ability to think critically about health, may potentially improve decision making in a large segment of the population and future health outcomes.

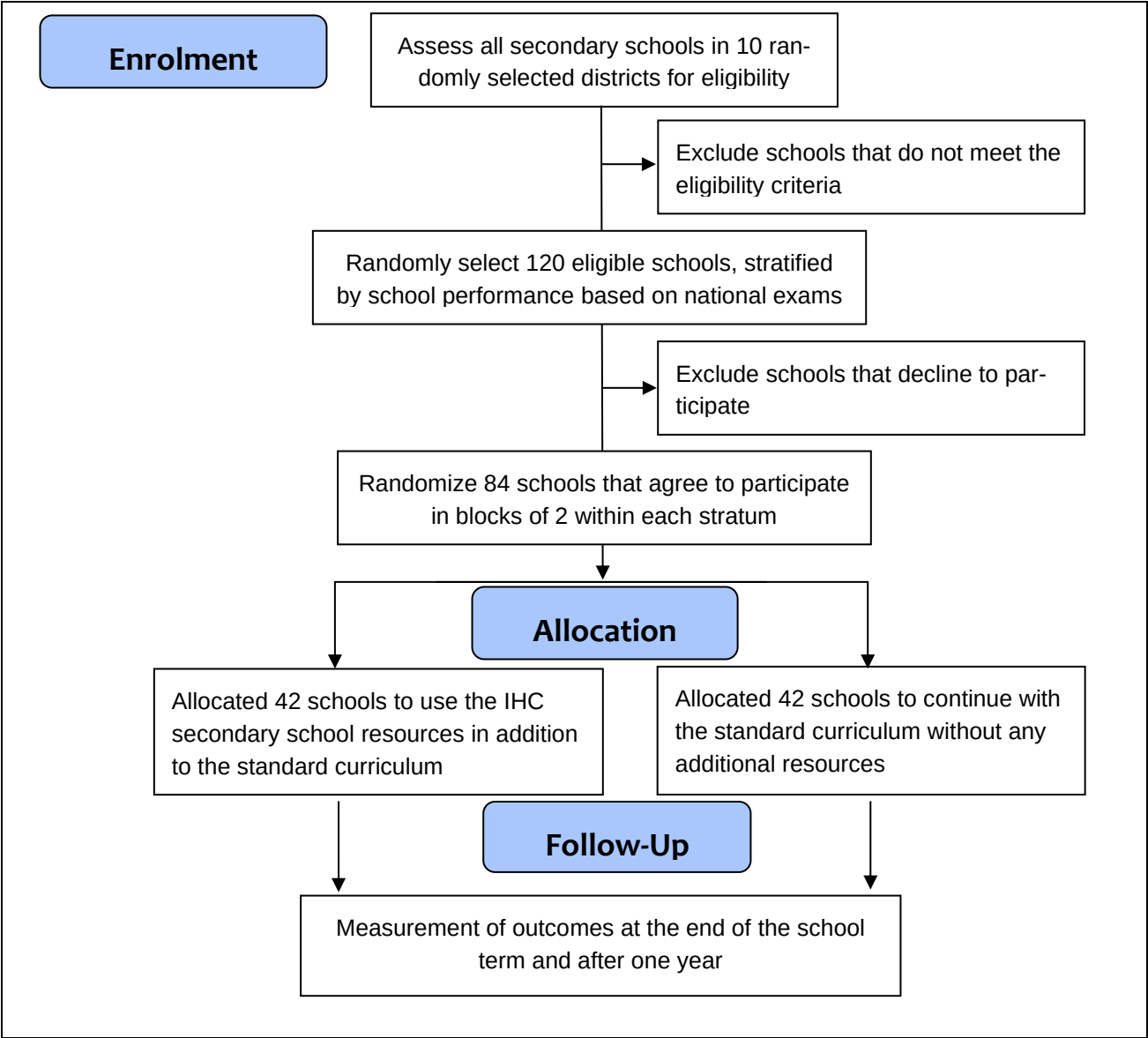
Despite these compelling reasons for teaching young people to think critically about health, few learning resources are available and hardly any have been rigorously evaluated [3, 4]. We have designed digital secondary school resources to help fill this gap and will evaluate the effects of using the resources in a randomised trial in Rwanda. Parallel randomised trials of using these resources will be conducted in Kenya and Uganda.

Methods

Design

We will conduct a two-arm, pragmatic cluster-randomized trial as illustrated in the flow diagram (Figure 1). We will use stratified random allocation to assign Rwandan secondary schools to intervention arm which will use the IHC secondary school resources in addition to standard curriculum. Schools in the control arm will continue with the standard curriculum with no additional resources.

Figure 1. Flow diagram



Research question

Does using the IHC secondary school (“*Be Smart about Your Health*”) resources in Rwandan secondary schools improve students’ ability to think critically about health compared to the standard curriculum?

Objectives

Primary: To evaluate the effect of the IHC secondary school resources on the ability of students in Rwanda to think critically about health.

Secondary: To evaluate effects of the IHC secondary school resources on students’ mastery of the nine key concepts covered by the resources, their understanding of and ability to apply each of the nine concepts, retention of what they learned after one year, intended behaviours, self-efficacy, and academic achievement. In addition, we will measure the effect of using *Be Smart about Your Health* on the ability of teachers to think critically about health.

Study setting and location

Rwanda is divided into five provinces and 30 districts. The districts govern primary and secondary schools. The present study will be conducted in lower secondary schools randomly selected from 10 districts representing all five provinces. Rwanda’s secondary school comprise six levels, from senior one to six (S1-S6). The first three years of secondary education are lower secondary (S1-S3). The official age range for secondary schools is 13-18 years. For lower secondary the official age is from 13-15 years. Students join secondary schools after successful completion of six years of primary school and having passed the primary national leaving examination. According to the 2019 education statistics [19], Rwanda has 1,783 secondary schools of which 1148 schools provide lower secondary education. Among 1783 secondary schools, 547 are publicly funded, 912 are government-aided (funded by government and private), and 324 are funded privately. In 2019 there were 481,138 students in lower secondary school, of which 34.2% were in senior-two. Overall, 54.3% of lower secondary school students were females. The average number of students per classroom in secondary schools was 39. Overall, there were 23,585 teaching staff in secondary schools. The average teacher to student ratio was 1 to 24.

As of 2019, most secondary schools had computers (85.5%) and grid electricity supply (76.6%). Two thirds (66.6%) had smart classrooms. Smart classrooms include student computers, projectors, smart boards, and Internet access. The ratio of computers to students is 1 to 8.

Lower secondary schools in Rwanda teach core subjects including mathematics, physics, chemistry, biology and health sciences, information and communication technology (ICT), English, French, Kinyarwanda, Swahili, history and citizenship, geography and environment, and entrepreneurship [20]. Lower secondary students have 41 periods in a week to cover all core subjects. Each period is 40 minutes. All subjects include generic competencies for higher-order thinking. The generic competencies in the curriculum are critical thinking, creativity and innovation, research and problem solving, communication, cooperation, interpersonal relations, life skills and lifelong learning [20].

Study population

The population for this trial will be S2 students and their teachers in public, private and government-aided lower secondary schools in Rwanda. S2 is the middle year in lower secondary school. Most S2 students can be followed up at the same school after a year, before graduating or dropping out. We will target teachers and students in the following subjects, where critical thinking about health would best fit into the curriculum: biology and health sciences, physics, chemistry, and mathematics [5].

Inclusion and exclusion criteria

The inclusion and exclusion criteria are listed in Table 1.

Table 1. Inclusion and exclusion criteria

	Inclusion criteria	Exclusion criteria
Schools	- Publicly funded, private funded or government-aided school - Schools with electricity - Schools using the national competence-based curriculum - Schools with lower secondary school section - Schools with a computer and Internet connection - Schools with over 100 students - Schools with over 10 teachers	- Schools that participated in the user testing and pilot of digital resources - International schools - Schools that provide special needs education - Schools that are hard to reach for geographical reasons
Students	Senior-two students	S2 students who opt not to attend the lessons
Teachers	- Teachers teaching one of the following subjects: biology and health sciences, physics, chemistry, and mathematics - Teachers who have access to a smartphone or computer	Teachers who do not provide informed consent

Sampling strategy

We will use multi-stage sampling to select schools. First, we will randomly select ten districts, two from each of the five provinces of Rwanda. We will contact the District Director of Education in each district through the Rwanda Basic Education Board (REB). We will work with the District Directors of Education to determine all eligible schools. Based on the Rwanda National Examination and School Inspection Authority data, eligible schools will be stratified by school performance. We will then randomly select schools in each district and stratify them by performance. The number of schools will be proportionate to the number of eligible schools in each district.

Recruitment strategy

We will work with district Directors of Education and REB to invite school directors (headmasters) from selected schools to attend an introductory meeting. At first, we will invite schools equal to the target sample size ($n=84$), and request headteachers to participate in the trial. Headteachers and teachers from schools that agree to participate in the study will sign consent forms. For schools that decline to participate, we will replace them with other schools similar in characteristics and invite them to participate in the trial. We will ask the school directors to select one senior-two class and identify one teacher. We will seek informed consent from the selected teachers for those classes. All students who agree to participate and sign an assent form will be included in the trial.

Random allocation

After recruitment, we will randomly allocate schools in the intervention or control arm using a computer-generated sequence. We will use block randomisation to ensure balance of key school characteristic that may influence outcomes (performance). We will use block sizes of six and four with equal numbers allocated to each arm. A statistician that will not be involved in recruitment of schools will generate the allocation sequence and assign schools to the intervention or control group. We will provide the statistician with a list of schools within each block, using a unique identifier for each school. The statistician will then prepare the randomisation list with the unique code and corresponding allocation group for each participating school. The list will not be changed after random allocation by the statistician.

Blinding

Due to the nature of the intervention, we cannot blind the trial participants or investigators. We will inform students and teachers about the trial and explain

to them its purpose, how they will participate, and the types of data we will collect from them. We will inform study participants how they scored on the outcome assessment tool (the number of correct answers out of 18 multiple-choice questions).

Sample size

We used the University of Aberdeen Health Services Research Unit's Cluster Sample Size Calculator [21] to estimate the sample size with the following assumptions:

- The number of students per cluster is 39, based on Rwanda education statistics for 2019 [19]
- The intraclass correlation coefficient (ICC) is 0.19, based on a cluster-randomised trial of primary school resources in Uganda [22]
- The proportion of students expected to achieve a passing score without the intervention is 30%, based on findings from the Ugandan primary school study [22]
- The smallest difference between the intervention and control arms we want to be able to detect is 20%, based on at least 50% of the students having a passing score in the intervention arm
- Alpha is 0.01
- Power is 90%

Based on these assumptions, we will need a minimum of 76 schools to participate in the trial. Allowing for up to 10% of schools not administering the outcome measure, we will recruit a minimum of 84 schools.

The intervention

A detailed description of the intervention is provided using the GREET checklist for describing educational interventions (Additional file 1). Schools allocated to the intervention will use *Be Smart about Your Health*. We have developed the resources iteratively using a human-centred design approach [23], working with students, teachers, and curriculum developers in Rwanda, Uganda, and Kenya [12]. The resources cover nine key concepts that people need to understand and apply to critically think about health choices (Box 1) [24, 25]. The secondary school resources include 10 lessons to be delivered to students in a classroom setting during a single school term (Table 2). Each lesson is designed to be delivered during a single 40-minute period.

Table 2. Outline of ten lessons with corresponding learning goals

	Lesson titles	Goals By the end of the lesson, students should be able to:
1	Health actions	• Identify health actions • Explain why it is important to think critically about health actions (why these lessons are important)
2	Health claims	• Identify claims about the effects of health actions
3	Unreliable claims	• Identify claims about the effects of health actions that are only based on personal experiences, how commonly-used something is, or how new or expensive something is • Explain why most such claims are unreliable
4	Reliable claims	• Explain why knowledge about the effects of health actions depends on comparisons • Explain why we need researchers to make the comparisons
5	Using what we learned (1)	• Remember what they learned in Lessons 1 to 4 • Use what they learned in these lessons in their daily lives • Recognise limits to what they have learned
6	Randomly created groups	• Explain why groups of people in a comparison should be similar at the start
7	Large-enough groups	• Explain what it means for comparisons between health actions to be large enough.
8	Personal choices	• Identify advantages and disadvantages of health actions, for individuals
9	Community choices	• Identify advantages and disadvantages of health actions, for communities
10	Using what we learned (2)	• Remember what they learned in Lessons 1 to 9 • Use what they learned in these lessons in their daily lives • Recognise limits to what they have learned

Teachers will be allowed to modify the delivery of the lesson to fit their context. We will include all participants in the analysis regardless of how the lessons were delivered. In addition to the lessons, teachers will continue to teach other subjects as usual and there will be no restriction on what is taught or how it is delivered. We will contact participating schools in the intervention arm before the start of the school term and invite all participating teachers to attend a three-day workshop (Additional file 2). The workshop will be at least one week before the start of the intervention delivery. In the workshop, we will explain to the teachers why critical thinking about health is important, the trial process, and expectations from schools and teachers. We will provide an overview of the 10 lessons.

Schools in the control arm will not receive the resources or training or be asked to teach anything new. We will meet with school directors in the control group to explain the trial and ensure that they are prepared for the test that will be used as the primary outcome measure at the end of the term. We will provide schools in the control arm with *Be Smart about Your Health* after the one-year follow-up, including the training materials.

Outcome assessment

Assessment tool

We will use the *Critical Thinking about Health Test* as an evaluation tool to measure the ability of students to think critically about health (Additional file 3). The test includes two multiple-choice questions for each of the nine key concepts covered by the secondary school resources. The multiple-choice questions are from the Claim Evaluation Tools item bank [26, 27]. Each question in the database begins with a scenario which is intended to be relevant across contexts, and which can be used for children (from age 10 and above), adult members of the public, and health professionals. People with expertise in research methods judged the items to have face validity, and end-users judged them relevant and acceptable in their settings [26]. We conducted cognitive interviews with secondary school students in Rwanda, Uganda, and Kenya, and we simplified some text, conducted pilot of the tool, and used Rasch analysis to assess the validity and reliability of the test (Additional file 4). We used a combination of Nedelsky's and Angoff's methods for setting performance standards on educational tests to determine cut-offs for passing and mastery scores (Additional file 5). In addition to the 18 multiple-choice questions, the test includes questions about intended behaviours and self-efficacy with Likert response options. The test will be administered to students and teachers at the same time.

Outcomes

The primary outcome measure is the difference between the intervention and control schools in the proportion of children with a passing score of at least 9 out of 18 correct answers (Additional file 5).

Secondary outcome measures are:

- The mean difference in the score (percent of correct answers) for the 18 multiple-choice questions
- The difference in the proportion of students with a score indicating mastery of the concepts covered by *Be Smart about Your Health* – at least 14 out of 18 correct answers (Additional file 5)
- The difference in the proportion of students that answered both questions correctly for each of the nine concepts covered by *Be Smart about Your Health*
- The difference in intended behaviours and self-efficacy among students based on the preset questions in the *Critical Thinking about Health Test*
- The difference in scores on standard national examinations
- The difference in the proportion of teachers with a passing or mastery score, and the mean difference in the score for teachers

All the outcomes will be measured at the end of the school term during which the IHC lessons are taught in the intervention schools and again one year later. Students and teachers in both study arms will receive their test scores.

Data collection and management

The *Critical Thinking about Health Test* will be administered in both the intervention and control schools by research assistants. The questionnaire will be administered electronically using Nettskjema, a service provided by University of Oslo [28]. If the test cannot be accessed online, for example because of a power outage or poor Internet connection, a paper version of the questionnaire will be used and the answer sheets will be scanned using ZipGrade [29]. The research assistants will guide the students on how to access test on the computer or, if necessary, how to use the answer sheets.

Individual students and teachers will be assigned a study code and be instructed to put their code on the questionnaire, and to answer all questions. The age and sex of each student (not teachers) will be collected using the questionnaire. Students and teachers in the intervention arm will complete the questionnaire after the intervention has been delivered. Those in the control arm will do so near to the end of the term during which the intervention is delivered. All participants from both arms will complete the same questionnaire one year after intervention delivery.

The research assistants will use school and teacher profile forms to collect descriptive information about the teachers and schools (Additional file 6). Before conducting the analysis, we will deidentify all data in the study database. The database will be kept on a secure server that can be accessed only by the study investigators. The study investigators will review the completeness of data and instruct the research assistants to collect missing or incomplete data when possible. We will supervise all the field work including data collection and entry processes. At the end of the study, the database will be closed, and the data will be cleaned and prepared for analysis.

Data analysis

We will use descriptive statistics and generalised linear models to analyse the data. We will use descriptive statistics to calculate the frequencies, percentages, mean and standard deviation for the baseline characteristics. We will use logistic regression for dichotomous outcomes and linear regression for continuous outcomes. We will use mixed models with a random effects term for the clusters (schools) and the stratification variables modelled as fixed effects. For the questions that assess applied knowledge or understanding, missing values

will be counted as wrong answers. We will report proportions, odds ratios, adjusted differences (based on the odds ratios), and the corresponding confidence intervals for binary outcomes. We will report the means, standard deviations, mean difference, and corresponding confidence interval for continuous outcomes. We will report the intraclass correlation coefficient for all student outcomes.

Sensitivity analyses

We will conduct two sensitivity analyses to explore the risk of bias due to attrition. First, we will conduct a weighted analysis for the primary outcome using inverse probability weighting. In this analysis, the students in each school will be given a weight equal to the inverse of the proportion of students in the school that completed the test. Second, to explore the risk of bias due to differential loss to follow-up between intervention and control schools, we will calculate upper and lower bounds for the mean difference in test scores using the Lee bounds approach [28]. These are constructed by trimming the group with less attrition at the upper and lower tails of the outcome (test score) distribution respectively. We will not adjust for covariates in this analysis.

Subgroup analyses

We will conduct two subgroup analyses. Firstly, we will compare the results for the primary outcome measure in schools with high and low performance as determined by Rwanda National Examination and School Inspection Authority. Secondly, we will conduct a subgroup analysis for students' English proficiency, based on responses to four questions testing their English proficiency in the questionnaire. We hypothesize that on average, students in high-performing schools will score higher than students in low-performing schools. We will conduct tests for interaction for the primary outcome for the subgroup analysis, and we will use published guidelines to interpret the results of the subgroup analysis [29, 30].

Trial management

This is one of three trials testing the effects of using the informed health choices resources in Rwanda, Kenya, and Uganda. The Norwegian Institute of Public Health, recipient of the grant from the Research Council of Norway, is the coordinating centre for the trials. The principal investigator for the trial in Rwanda is MM. The principal members of the coordinating group for the trial are LN, MK, NKS, SR and AO. The steering committee for all three trials includes principal members (LN, MK, NKS, SR and AO) and principal investigators (MM, FC and RS). They are responsible for final decisions about the protocol and reporting of the results. The principal investigator of this trial in Rwanda is responsible for

managing fieldwork, overseeing recruitment, data collection, data management and publication.

We will keep all study documents at the University of Rwanda in a locked cupboard. The documents will include the protocol, ethics approvals, collaboration letters, consents, data collection forms for participants, and the data management plan. We will create a separate folder for each school with the school code, students' and teachers' names and codes, the intervention delivery plan, and any adverse events recorded at the school. The documents will be kept for five years, after which they will be destroyed.

The timeline for the trial is shown below (Table 3). Recruitment of schools will begin in March 2022, the *Be Smart about Your Health* lessons will be taught in the intervention schools between April and July 2022. The outcome assessment will be conducted between June and July 2022 and the one year follow up will be conducted from May 2023.

Post recruitment retention strategies

When the school directors give their consent for their school to participate (Additional file 7), they will agree to allocate the time for teaching the IHC lessons and administering the *Critical Thinking about Health Test*. In addition, we will obtain consent from teachers in the intervention arm prior to teacher training. We will obtain consent of teachers in the control arm through their school directors. At recruitment meetings with the school directors and the introductory meetings with teachers, we will explain the purpose of the trial and the importance of completing the lessons and test. The lessons are designed to engage both teachers and students. They have been piloted and user-tested to ensure that users experience the resources as useful, easy to use, and well-suited to the context. At the beginning of each lesson, classes will review the previous lesson. At the end of each lesson, they will preview the next lesson. We will call teachers in participating schools during the school term, to know when the lessons will be taught or when the test will be administered for control schools. We will call teachers again during the time between the first and second outcome assessment to remind them of the trial's purpose and the importance of students in both the intervention and control schools completing the test. Targeting senior-two students will facilitate data collection after a year, before students are allocated to other schools after the lower secondary school national examination.

Ethical considerations and informed consent

Ethical conduct of the study

The study will be performed following the protocol, regulatory requirements, guidelines, and principles for conducting studies involving human subjects in Rwanda.

Ethical approval

We sought ethical clearance from the Rwandan National Ethics Committee (RNEC) prior to the start of the project in August 2019 and the protocol has been reviewed annually by the RNEC since then. We will seek approval from the RNEC for any changes to the protocol before they are implemented.

Consent and assent

The head of each participating school will provide written or electronic informed consent for the school and participating students. A list of participating students (Additional file 8) will be attached to the consent forms. Participating teachers and students with 18 years and above will provide written or electronic informed consent for their participation (Additional file 9), and

participating students below 18 years will provide informed assent (Additional file 10).

Safety monitoring and adverse events

This study poses very low risk to participants. However, researchers often overlook the potential adverse effects of educational [27] and public health interventions [28-29]. To minimise and evaluate potential adverse effects of the IHC digital secondary school resources, we are developing a framework with theoretical mechanisms (Additional file 11). We will ask teachers in the intervention arm to record potential adverse effects and report them to the investigator (MM). At the start of the trial, we will give instructions for recording and reporting adverse events (Additional file 12).

Stakeholder engagement

We have engaged key stakeholders in developing the intervention and throughout this project [12]. We will engage education authorities in recruiting participating schools. We have sought feedback on this protocol from the Rwanda Basic Education Board, members of our national advisory group, and teacher and student networks. Before publishing the trial results, we will discuss the findings and their interpretation with those stakeholders.

Reporting, notification, and dissemination of results

We will share the main findings of the trial with the schools through an email to school directors and the district directors of education. After the trial, we will conduct a workshop with key stakeholders in education and present key findings. We will submit reports of the results after the first outcome measure and the one-year follow-up to a relevant journal for publication and dissemination. The final de-identified dataset and analysis code will be deposited in an open access repository (Zenodo.org).

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Protocol contributors

Conceptualization, MM, NL, FC, SR, OM, NA, SD, MK, SN, SER and AO; Methodology, MM, NL, SCMC, FC, SR, OM, NA, SD, AA, MK, LS, SKN, SER and AO; Writing—original draft, MM; Writing—review & editing, All authors.

Competing interests

All authors declare they have no competing interests.

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Additional files

Additional file 1	GREET checklist describing the IHC secondary school intervention
Additional file 2	Teacher training workshop
Additional file 3	Critical Thinking about Health Test
Additional file 4	Development and evaluation of the Critical Thinking about Health Test
Additional file 5	Determination of cut-offs for passing and mastery scores
Additional file 6	Questionnaire for collecting school and teachers' characteristics
Additional file 7	Consent form that school directors will sign to allow schools to participate in the study
Additional file 8	Template for list of students who will participate in the study
Additional file 9	Consent form that teachers/adults will sign before their participation in the study
Additional file 10	Assent form that students will sign before their participation in the study
Additional file 11	Framework for preventing and evaluating potential adverse effects
Additional file 12	Form for reporting adverse events
Additional file 13	The standard protocol Item (SPIRIT) checklist